

Online Experiments for Language Scientists

Lecture 5: Frequency learning

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Adapted from slides by Kenny Smith

Reminder about Undergraduate Assessment 1

- Due on 6th November
- If you haven't already, read:
 - [The assignment brief](#)
 - [The FAQ](#)
 - [The generative AI policy](#)
- No questions after 4pm on Monday 3rd November!

Today's paper

Ferdinand, V., Kirby, S., & Smith, K. (2019).
The cognitive roots of regularization in
language. *Cognition*, 184, 53-68.

Large frequency-learning experiment run
on MTurk

- Do domain (linguistic vs non-linguistic)
and demand (tracking 1 vs 6 frequency
distributions) influence regularization
behaviour?



**Vanessa
Ferdinand**
(formerly
*Edinburgh, now
Melbourne*)



Simon Kirby
(*Edinburgh*)



Kenny Smith
(*Edinburgh*)

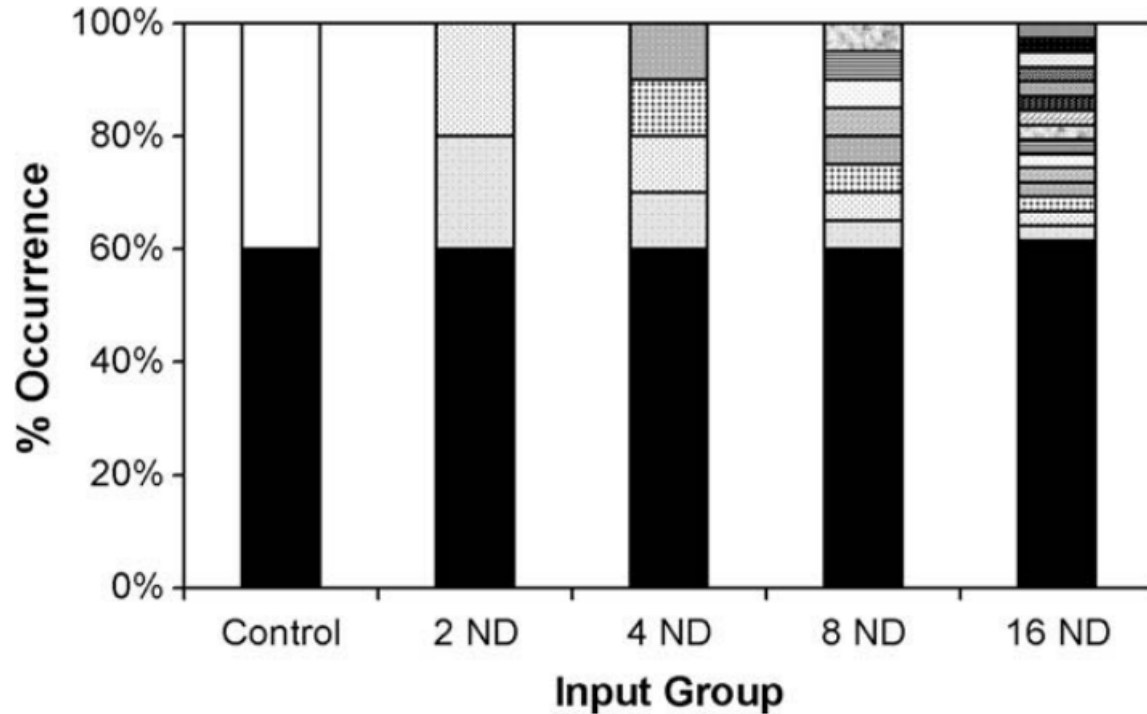
Variation in language

Languages exhibit variation at all levels (paraphrase, synonymy, allomorphy, allophony), but variation is **constrained**

- Languages have lexicons and grammars
- Linguistic (phonological, lexical, syntactic, semantic or sociolinguistic) **conditioning** of alternation
 - English past tense allomorphy: hunt/**ɪd**/ vs fish/**t**/
 - Noun classes: **la** chaise, **le** sofa, **la** fille, **le** garçon
 - T-glottaling: glo/**t**/al vs glo/**ʔ**/al

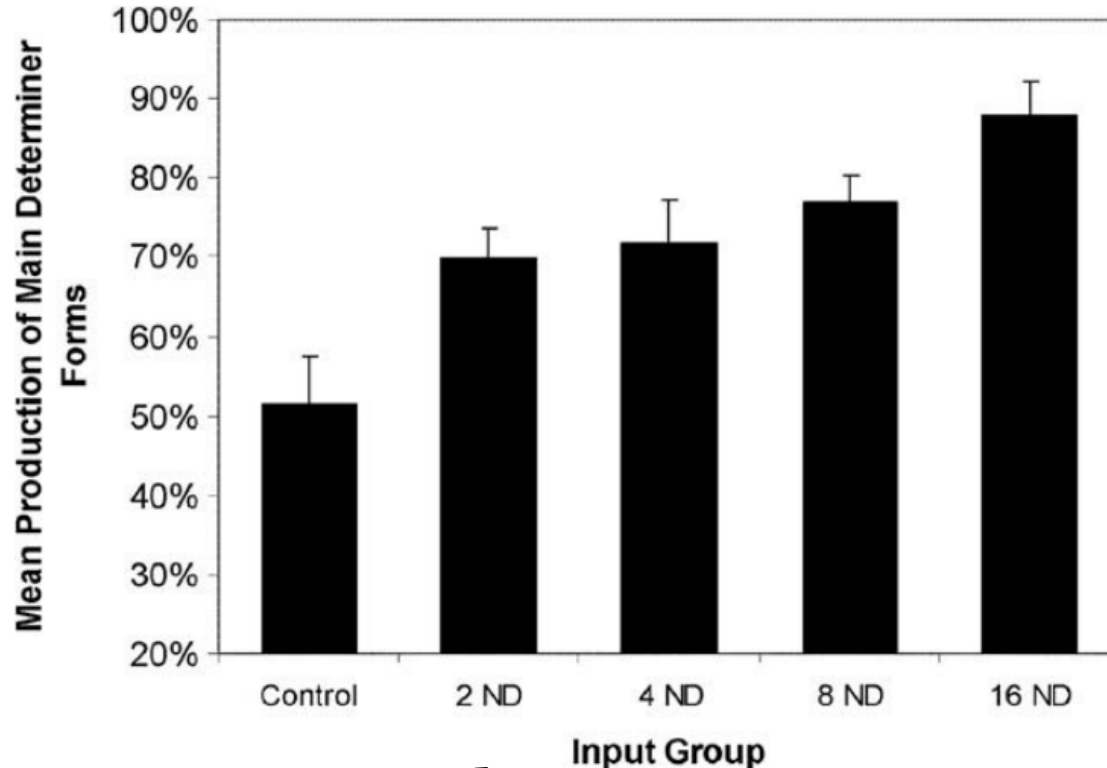
Why is language like this?

The possible role of language learners



Hudson, Kam & Newport, 2009, Experiment 1

The possible role of language learners



Hudson, Kam & Newport, 2009, Experiment 1

The possible role of language learners

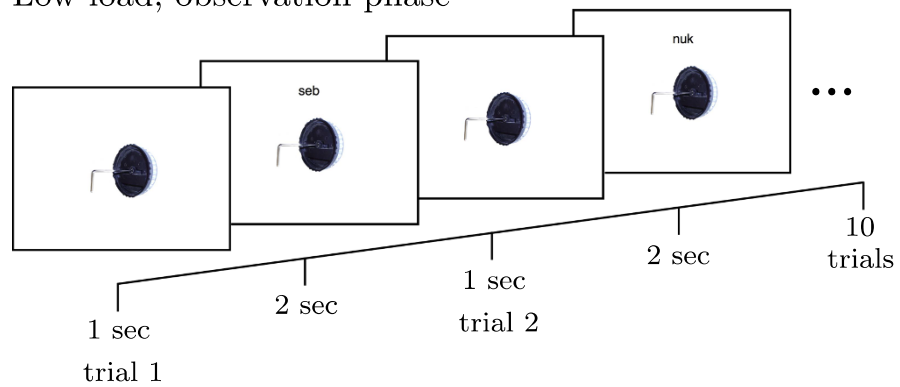
- Domain general or domain specific?
- Cognitive load?
- Encoding or production?

The present study

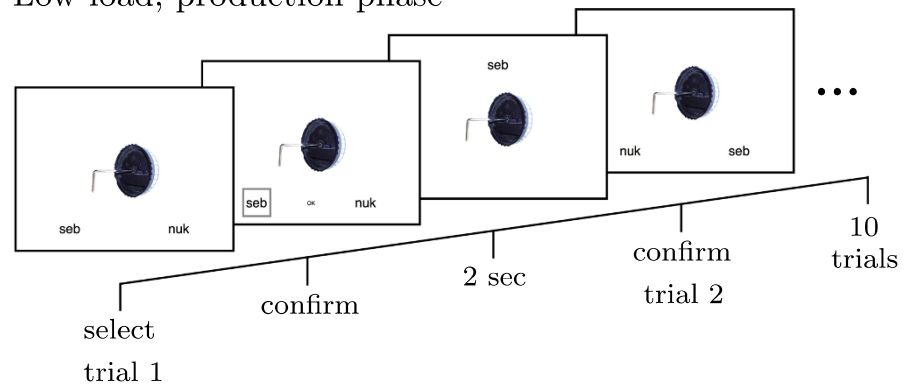
- Cognitive load: 1 distribution vs. 6 distributions
- Task domain: objects and words vs. containers and marbles

	Low load	High load
Linguistic	words1	words6
Non-linguistic	marbles1	marbles6

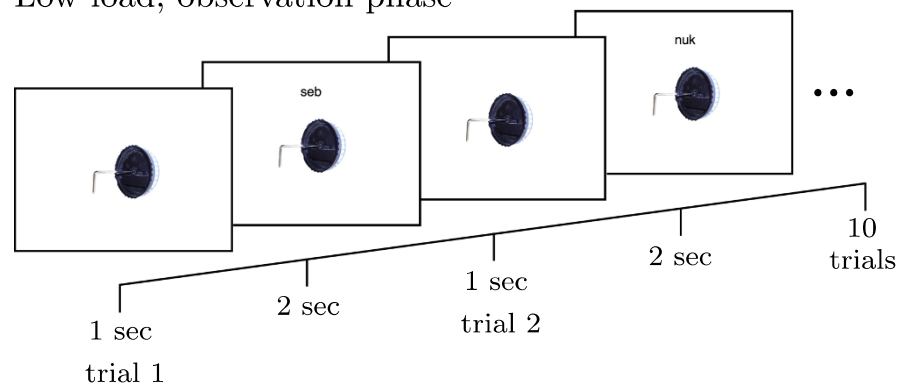
Low load, observation phase



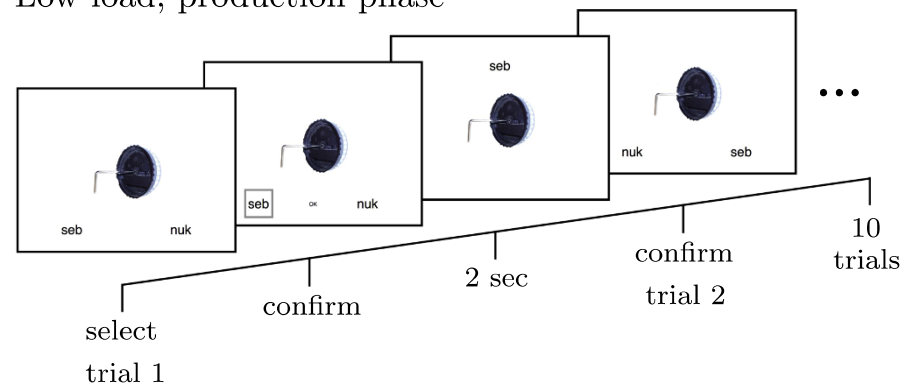
Low load, production phase



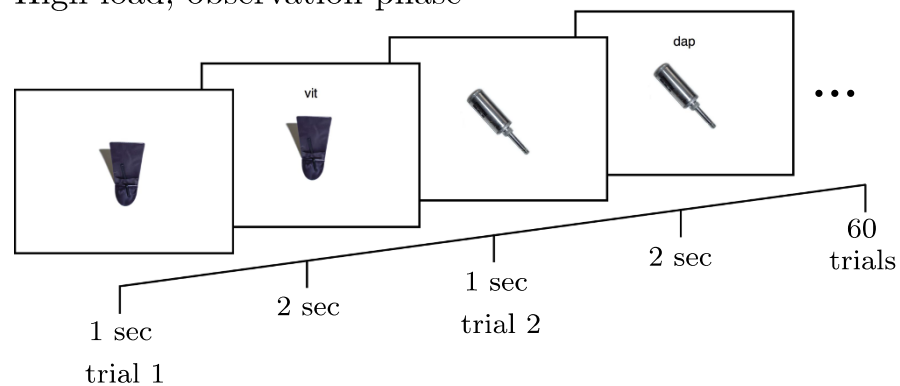
Low load, observation phase



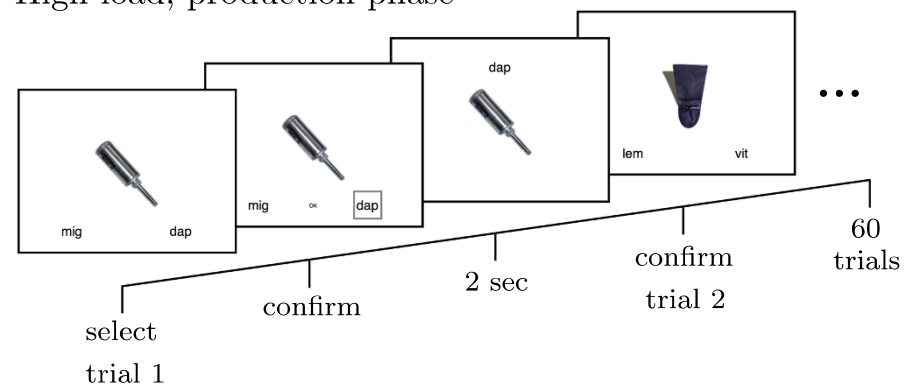
Low load, production phase

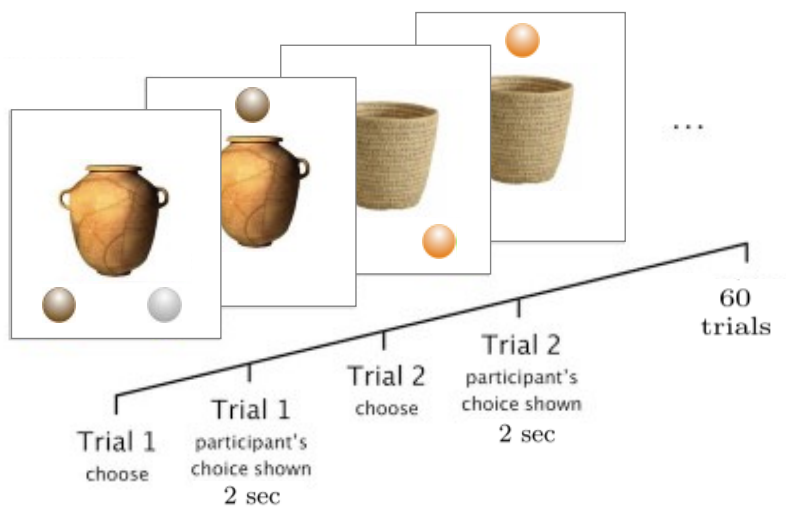
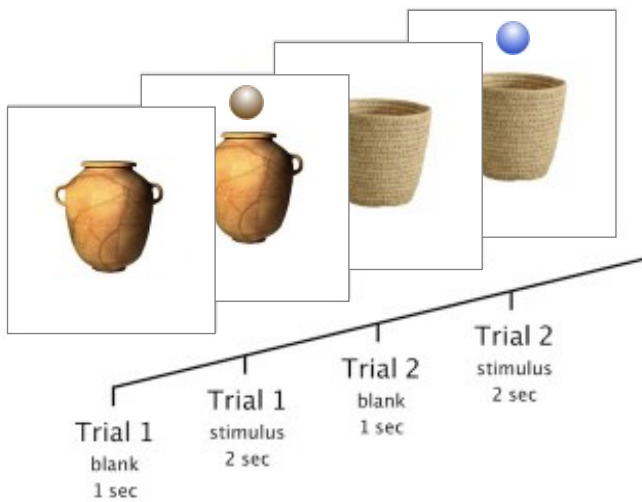


High load, observation phase



High load, production phase





Frequency learning experiment

Observation: observe a distribution

Production: produce a distribution



Variable of interest:
Difference between
these distributions

Estimation: estimate the frequencies of the input stimuli

- Simplified implementation: [this week's lab experiment](#)

How to measure regularization?

- **What we want to capture:** The amount of variation lost in learners' output compared to their input
- Entropy: A measure of variation
- **Method:** Calculate the reduction in entropy between input and output

Entropy

- Would it be easier to guess the outcome of rolling a dice or flipping a coin?



Entropy

- Would it be easier to guess the outcome of rolling a fair dice or an unfair dice?

Fair dice: 16.7% of getting '6'

Unfair dice: 90% of getting '6'

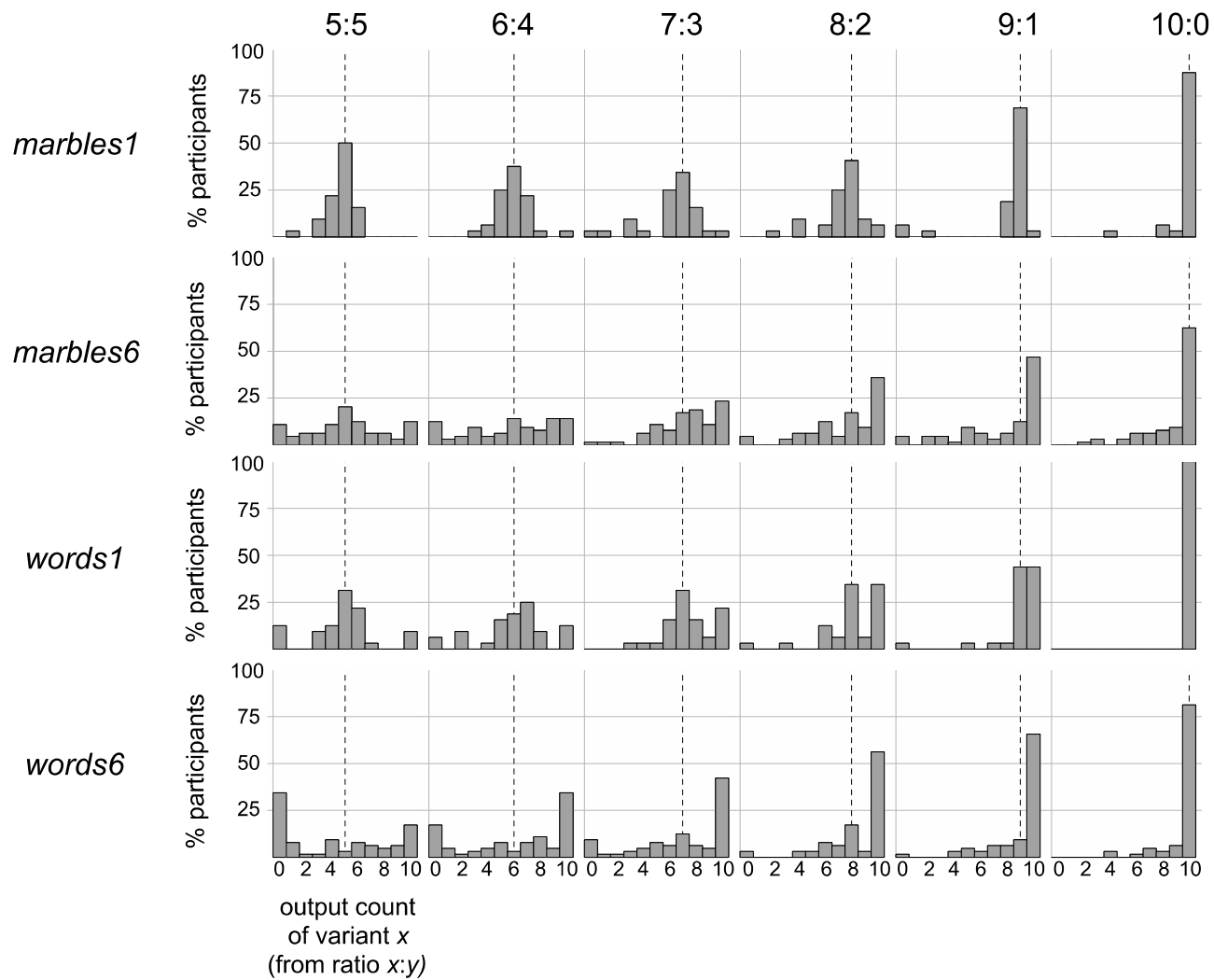


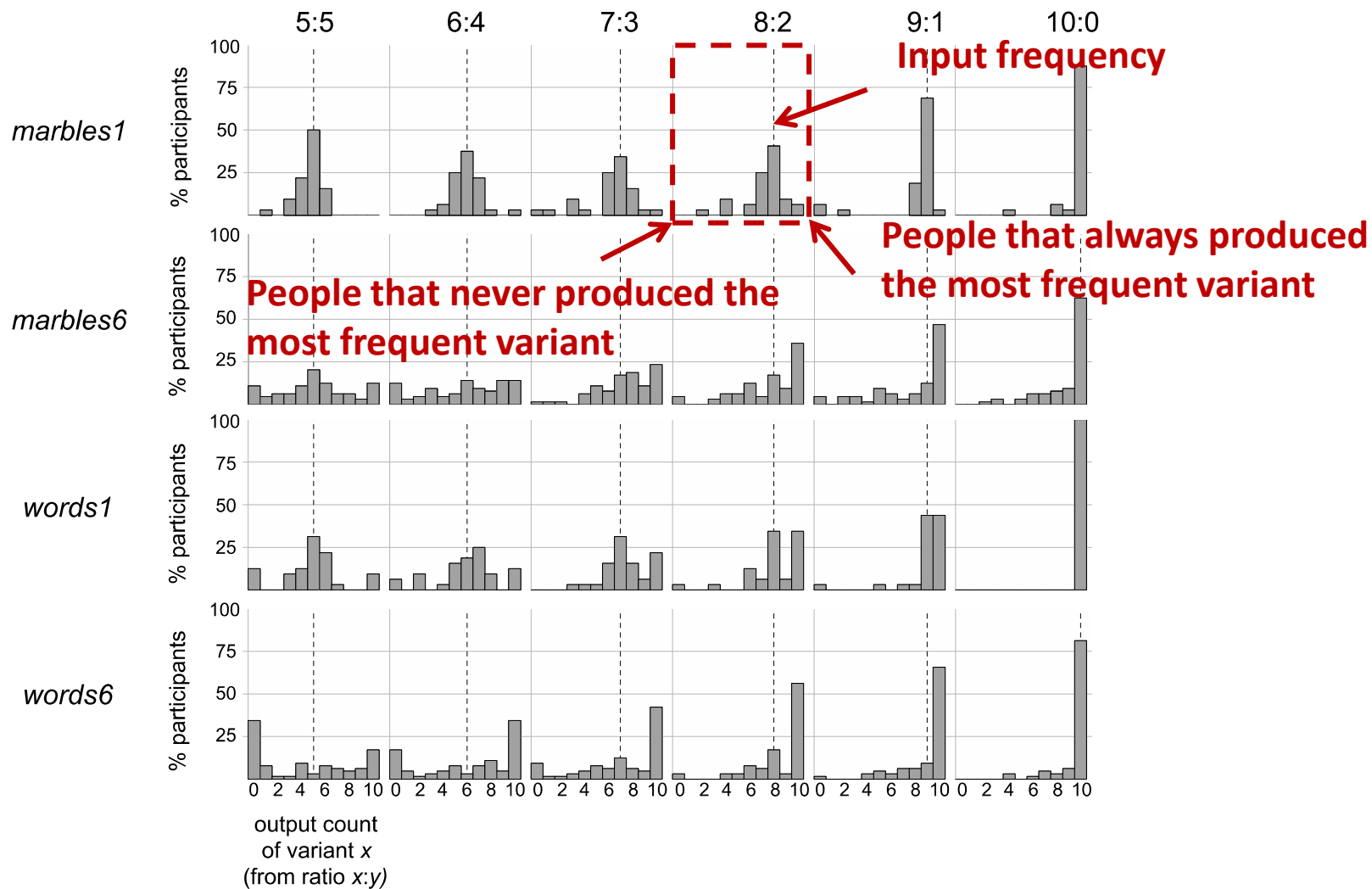
Entropy

- Events with more possible outcomes are more variable than those with fewer outcomes
- Events with evenly distributed outcomes are more variable than events with skewed distributions
- Entropy captures both: How variable is a distribution? (How hard would it be to guess what's coming next?)

Sample size, study duration etc

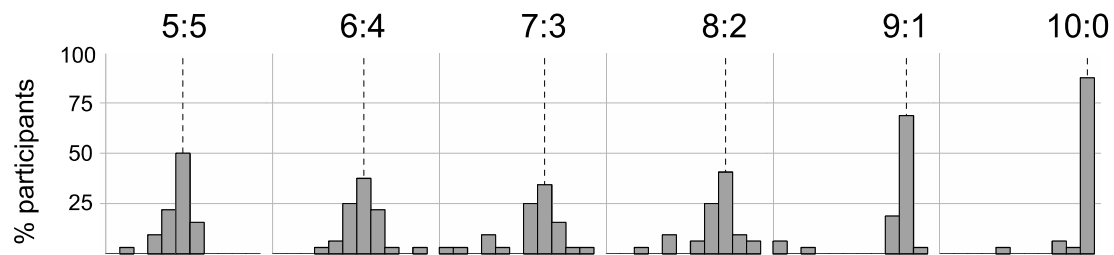
- US-based MTurk workers
- N=512 after exclusions
- 4 minutes (1-item task) or 11.5 minutes (6-item task)
- **\$0.10** (1-item task) or **\$0.60** (6-item task) 😞





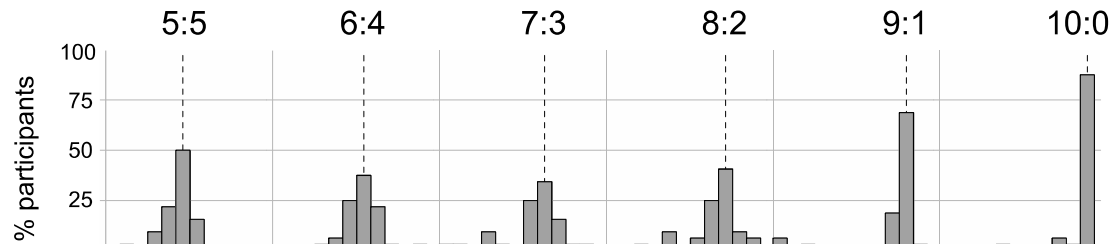
Probability matching

marbles1

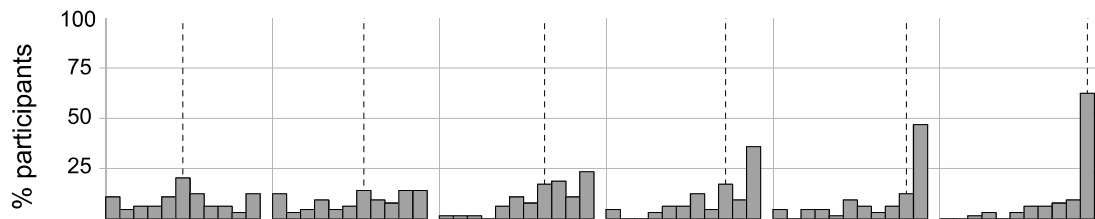


**Probability
matching**

marbles1



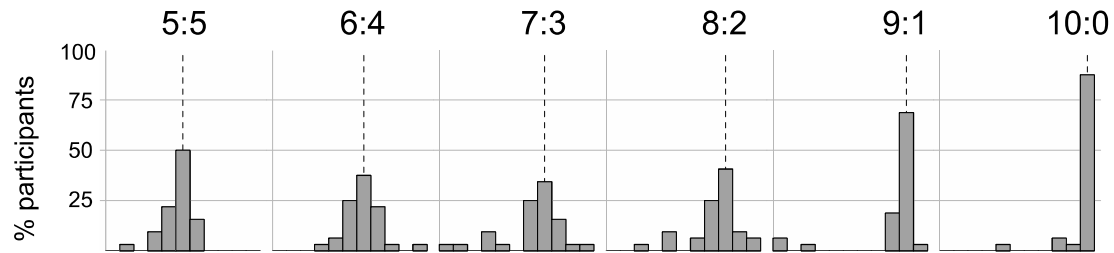
marbles6



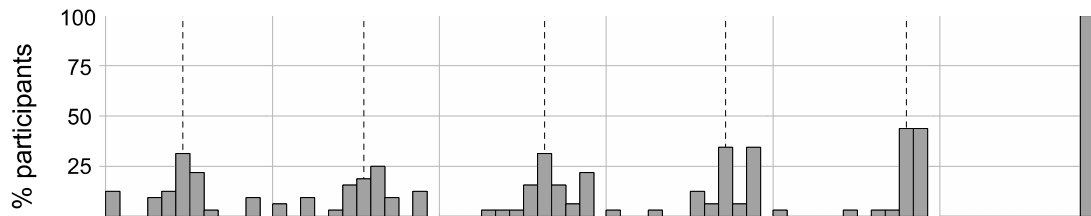
**Demand
effect**

**Probability
matching**

marbles1



words1



**Domain
effect**

5:5

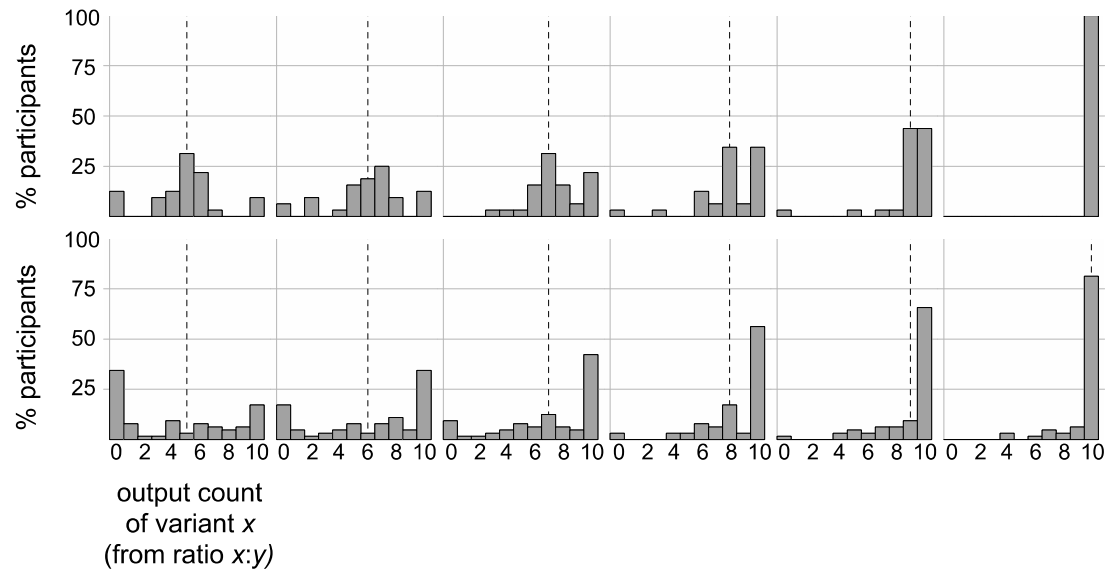
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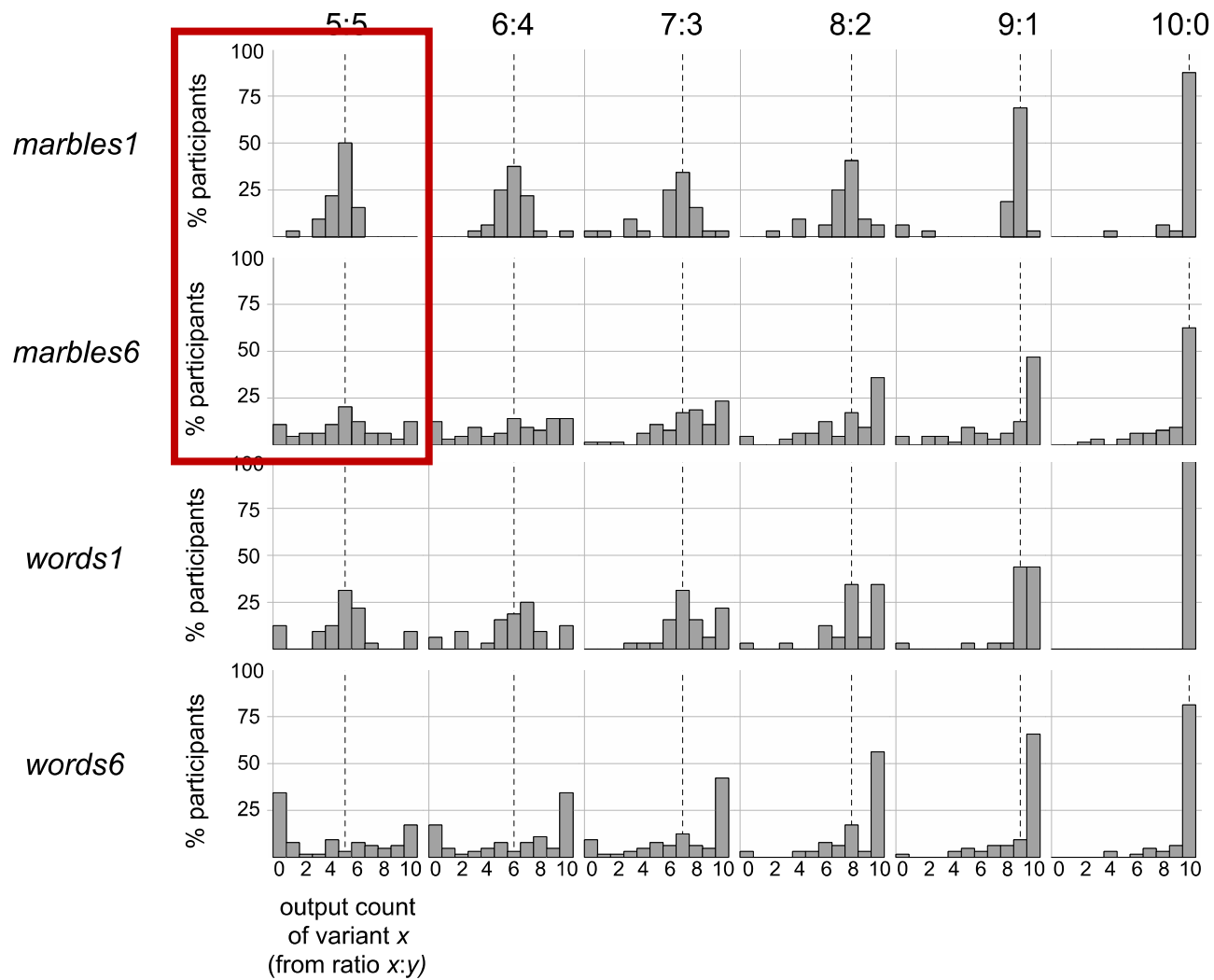
7:3

8:2

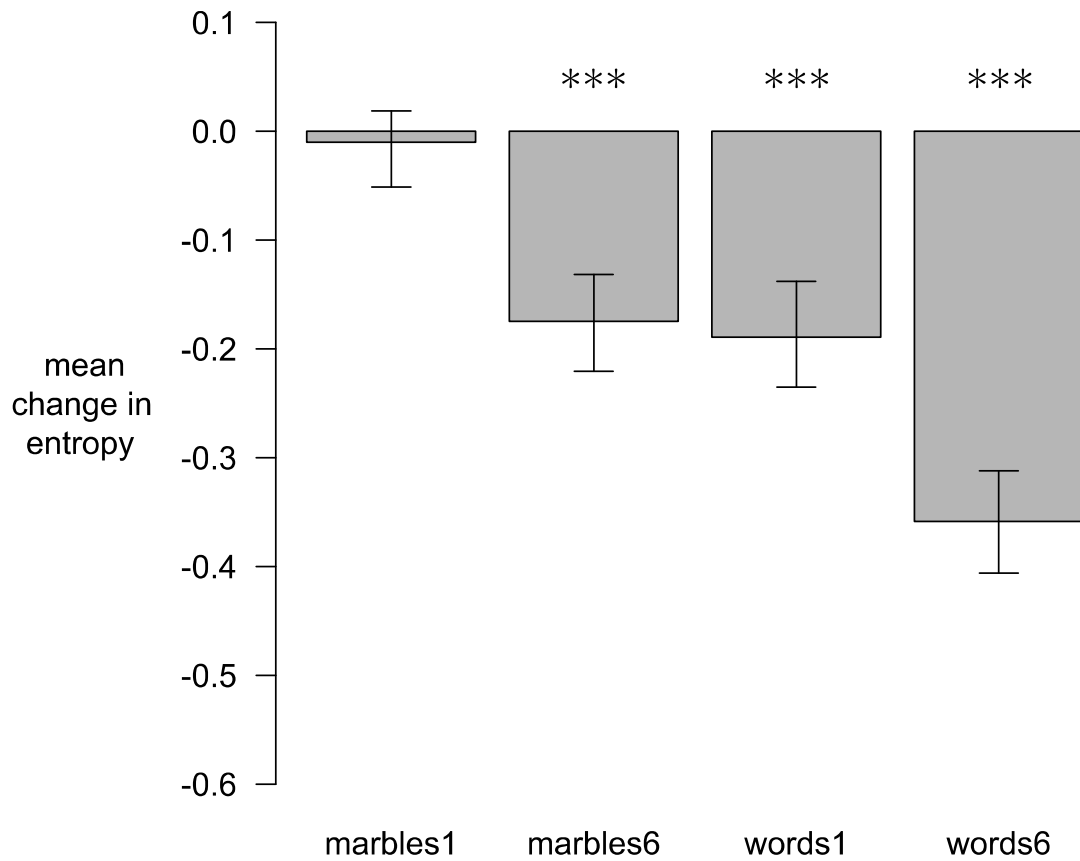
9:1

10:0

words1**Demand
effect**



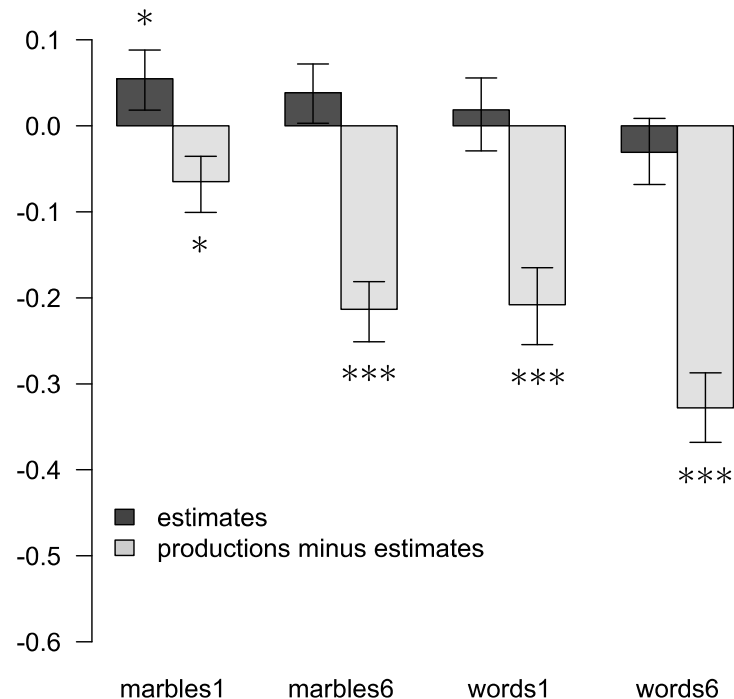
**How much
the output
is more
regular
compared
to the input**



Regularization during encoding or retrieval?

	tef	0	10	20	30	40	50	60	70	80	90	100
	gos	100	90	80	70	60	50	40	30	20	10	0
	fud	0	10	20	30	40	50	60	70	80	90	100
	pon	100	90	80	70	60	50	40	30	20	10	0
	seb	0	10	20	30	40	50	60	70	80	90	100
	nuk	100	90	80	70	60	50	40	30	20	10	0

mean
change in
entropy



Individual differences

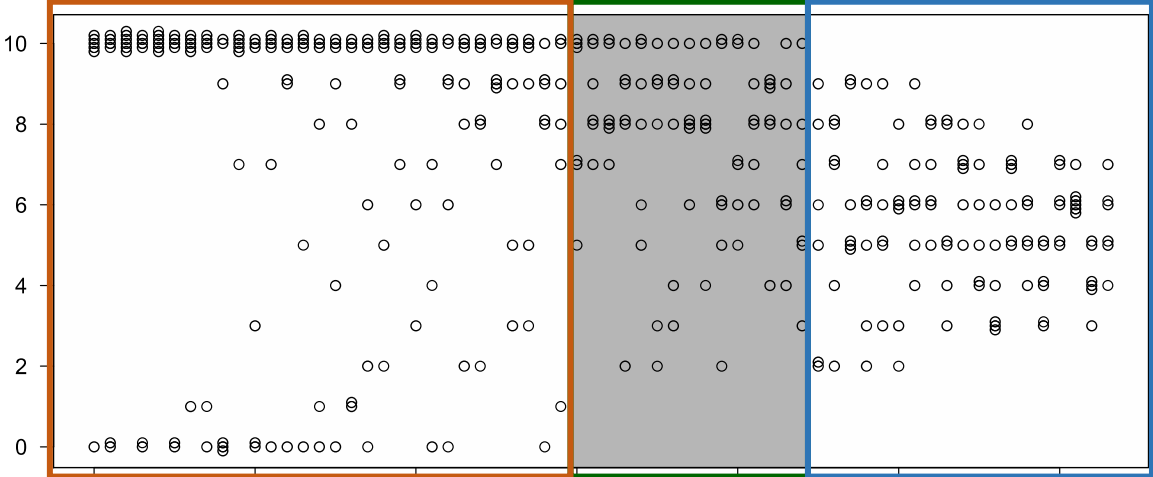
Probability matchers

Variabilizers

Regularizers

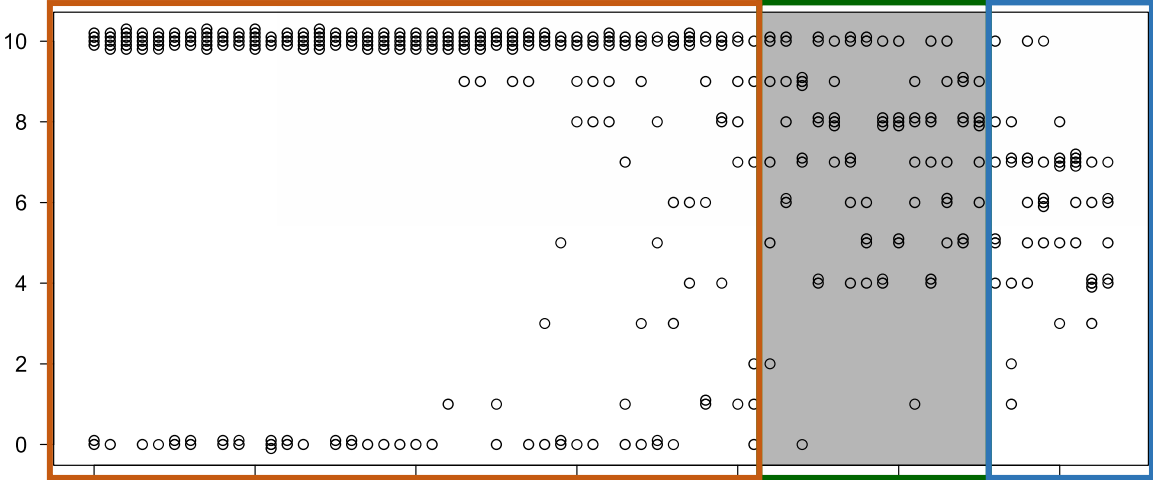
marbles6

output count
of variant x



words6

output count
of variant x

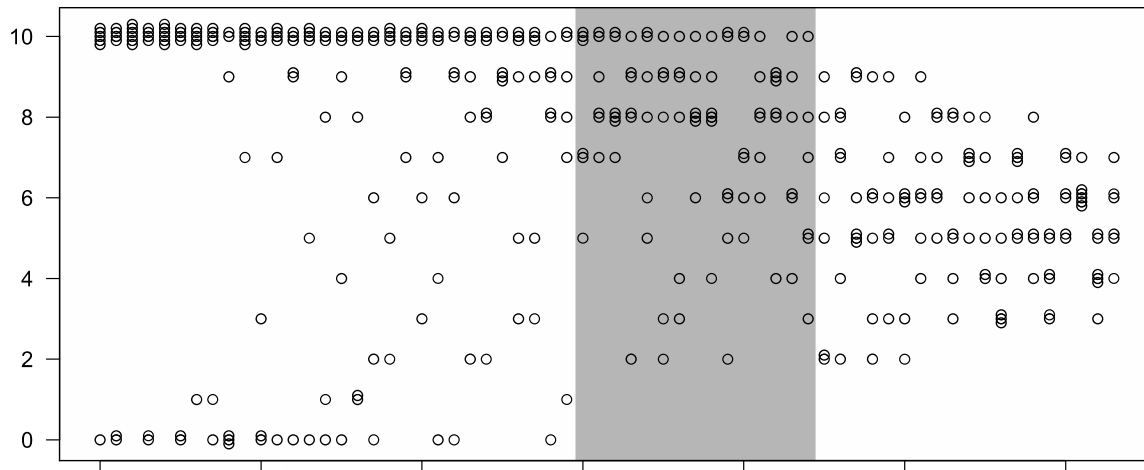


participant

Minority regularizers

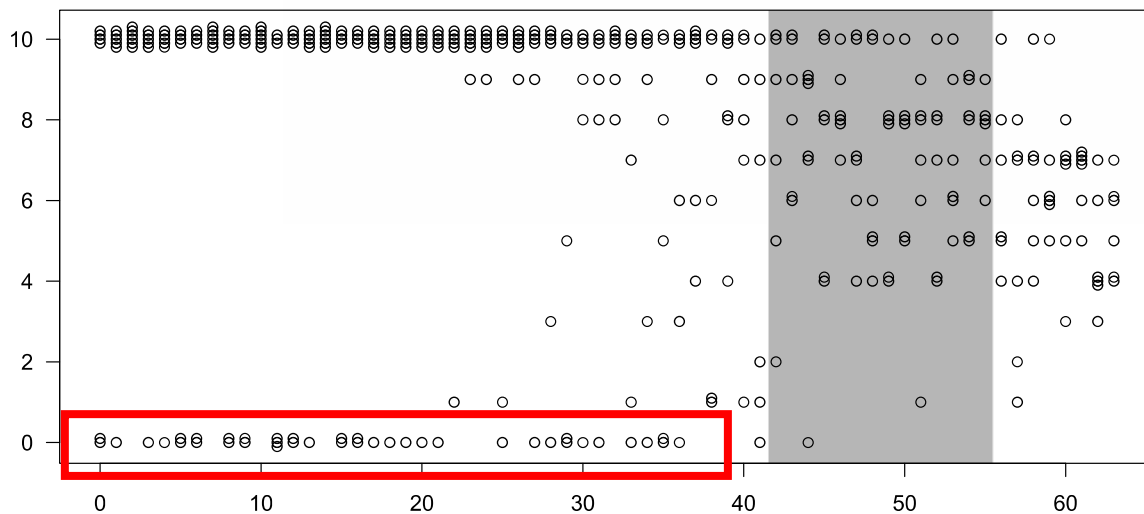
marbles6

output count
of variant x



words6

output count
of variant x



participant

Ferdinand et al.'s conclusions

- Effects of domain and demand on regularization
 - More regularization on linguistic than non-linguistic tasks (why?)
 - More regularization when under greater cognitive load
- Regularization effects mainly in retrieval (not encoding)

Time for Q&A/discussion on this
week's reading!

Next up

Lab

- A frequency learning experiment

Next week:

- Perceptual learning, audio stimuli
- The end of the “basics”